

Sem 4th DBMS Assignment Deadline Sept 1st Week 2026

- **Sec A** **3rd Sept**
- **Sec B** **4th Sept**
- **Sec C** **3rd Sept**

Enterprise Application Development Assignment (Python + Oracle Database)

Assignment

You are required to develop a Python-based Enterprise Web Application and web client with the functionality detailed below. The application that you develop must not use examples developed in tutorials, as you must choose your own original scenario.

Your server application will run using a Python web framework such as Django or Flask and will use Oracle Database as the backend.

The application should meet the following minimum requirements:

- An Oracle database consisting of three (3) or more linked tables.
- ORM model classes corresponding to the database tables.
- Data Transfer Objects (DTOs) or API serializers that can transfer data between the server and client.
- A business service layer that provides the application functionality.

Functionality that includes:

- The ability to create, retrieve, update, and delete records from each table.
- Three queries that navigate related database entities and return information (e.g., return all vehicles assigned to a particular driver).
- Two more complex queries involving at least three related database entities and returning meaningful information.
- A background task or asynchronous processing component that updates a database table based on a client request.

You are also required to develop the following client:

- A web-based graphical user interface (GUI) that uses the functionality provided by the server application.
- Report Requirements

You are also required to produce a report containing the following:

- Table descriptions.
- Entity-Relationship (E-R) Diagram.
- Class Diagram.
- Screenshots of the application running.
- The report must be submitted in Markdown (.md) format.
- Submission Requirements

You must submit the following on the deadline date:

- A zipped copy of the complete Python project, including both server-side and client-side components.
- Oracle Database SQL scripts required to create the database.

The project report.

- After submitting your report, you are also required to demonstrate the functionality of your enterprise application to the lecturer.
- Failure to demonstrate the application or failure to fulfil all submission requirements may result in the assignment being failed.

Development Guidelines

- Follow the standard software development process, including analysis, design, implementation, testing, and deployment.
- Begin by understanding the application requirements and designing the web pages, navigation structure, database entities, and application architecture.
- Divide the application into presentation, business, and data access layers.
- Clearly define the interfaces among these components before implementation.
- Implement and test each component independently before integrating the complete system.
- If changes are required during implementation, update the design documentation accordingly.

Before submission, test the complete application on another machine or a clean environment to ensure that the project deploys and functions correctly.

Software Requirements

- Python 3.x
- Django or Flask
- Oracle Database
- Oracle SQL Developer
- Visual Studio Code or PyCharm

- Git (recommended)
- Deliverables

The final deliverable consists of:

- Design document
- Source code
- Deployable application

The design document should describe the system architecture, components, and customized application framework.

Your design document should include the following contents:

- Cover page
- Table of contents (with page numbering)
- System proposal
- Design of web components with a brief description of the technologies used (Django Templates, Flask Templates, REST APIs, etc.)
- Web page design, including a navigation diagram of all pages
- Design of the business layer with a brief description of how business logic is implemented
- Design of the database access layer using an ORM (e.g., Django ORM or SQLAlchemy)

Overview of the application

- Project plan
- UML diagrams
- Description of the system architecture and interaction among the application layers
- Database design, including descriptions of each table and an Entity-Relationship (E-R) Diagram
- Also add User Manual and Application Installation Manual